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CHAPTER



Derivatives Markets in Developing Countries

We conclude this book by discussing the role of derivatives markets in developing countries.

Derivatives have become very important tools for transferring risks from one entity to another. It is not surprising that derivatives markets are growing fast in many developing countries. This chapter focuses on what is happening in China and India. These are two countries whose economies are expected to play dominant roles in the 21st century. China's population in 2009 is estimated to be about 1.3 billion, while that of India is about 1.2 billion. (By contrast, the population of the United States is only about 0.3 billion.) The world's population is about 6.8 billion, so China and India between them account for about 37% of the world's population. India is expected to overtake China as the world's most populous nation by 2030.

There can be little doubt that China and India will have a huge impact on the development of derivatives markets throughout the world in the years to come. Other countries will also be important players. For example, Brazil, the fifth most populous country in the world, has been very successful in developing its derivatives markets. Its premier exchange, BM&FBOVESPA (www.bmfbovespa.com.br) is highly regarded.

36.1 CHINA'S MARKETS

The way derivatives markets are regulated plays an important role in their growth. Too much regulation can stunt growth; not enough is liable to lead to a lack of confidence in the markets and make individuals and corporations less willing to trade.

In China, the China Securities Regulatory Commission (CSRC: www.csrc.gov.cn), the China Bank Regulatory Commission (CBRC: www.cbrc.gov.cn) and the People's Bank of China (PBC: www.pbc.gov.cn) are all involved in derivatives regulation. The CBRC issued new rules concerning derivatives in February 2004. This eased many of the restrictions that had previously been in place on the trading of derivatives. The new rules outlined the approval processes that would permit financial institutions to

trade derivatives and the risk management procedures that the institutions would be required to observe. The derivatives that can be traded include forwards, futures, swaps, options, and structured products. The underlyings include foreign exchange and interest rates. The rules allowed financial institutions to enter into derivatives transactions for both hedging and profit-making purposes. This was an important change for the market's potential for growth. Until 2004, financial institutions were allowed to enter into derivatives transactions only for hedging purposes. As explained in Chapter 1, the main reason why derivatives markets have grown so fast in the Western world is that they can be used for three different purposes: hedging, speculation, and arbitrage.

Prior to 2004, there were some mishaps. For example, Guangdong International Trust and Investment Corporation went bankrupt in 1998. It had been a major player in derivatives markets and structured products, but the legal foundation for what it was doing was questionable. During the bankruptcy proceedings, the Chinese courts took the view that the financial institution did not have the authority to enter into its contracts and declared all the contracts null and void. The 2004 rules give much clearer guidelines than before on who can trade derivatives and what they can be used for. The rules allow foreign banks to deal directly with Chinese entities, rather than by going through a Chinese financial institution.

China's official currency is the yuan (also known as the renminbi, RMB, or CNY). The Chinese government has indicated that one of its goals is to make the yuan fully convertible. Convertibility means that the exchange rate between the yuan and other major currencies such as the euro and the US dollar will be determined by supply and demand in the same way that other exchange rates between the major currencies of the world are determined. At present the government restricts movements in the exchange rate between the yuan and the US dollar from day to day and many economists think that the yuan is currently undervalued against the US dollar, with the result that China's exports have a competitive edge and China's trade surplus is boosted.

There are a number of exchanges in China that trade futures contracts. The Shanghai Futures Exchange (SHFE: www.shfe.com.cn) trades futures contracts on copper, aluminum, natural rubber, fuel oil, gold, zinc, and steel. The Dalian Commodity Exchange (DCE: www.dce.com.cn) trades futures contracts on soybean, soybean meal, soybean oil, corn, linear low-density polyethylene, polyvinyl chloride (PVC), and palm olein (which is a product obtained from the fractionation of palm oil). In 2008 it became the world's second largest agricultural futures exchange (after the Chicago Board of Trade). The Zhengzhou Commodity Exchange (ZCE: www.zce.cn) trades futures contracts on wheat, cotton, sugar, pure terephthalic acid (a chemical used in the manufacture of clothing and plastic bottles), canola, rice and green beans.

On September 8, 2006, several exchanges (including the three futures exchanges that have just been mentioned) founded the China Financial Futures Exchange (CFFEX: www.cffex.com.cn). In January 2010, regulators finally approved the launch of a CSI 300 index futures. The CSI index tracks the daily price performance of 300 stocks listed on the Shanghai Stock Exchange (SSE: www.sse.com.cn) or Shenzhen Stock Exchanges (SZSE: www.szse.cn). The index was launched in 2004 and was set equal to 1,000 on December 31, 2004. In China, as in many other countries, it has proved more difficult to obtain regulatory approval for the trading of index futures than for commodity futures. This is because they require cash settlement. Regulators are reluctant to approve contracts requiring cash settlement because they consider them

closer to gambling than contracts that are settled by physical delivery of the underlying. (Arguably this is not rational because, as explained in Chapter 2, the vast majority of futures contracts are closed out before maturity and therefore effectively settled in cash.) The trading of options on exchanges in China is not as well developed as the trading of futures.

The contracts traded in China's over-the-counter market include forward and swap contracts on foreign exchange, interest rate swaps, and bond forwards. Convertible bonds have been popular. For example, China's second-largest telecommunications equipment maker, ZTE, announced in January 2008 that it was offering 4 billion yuan (\$555 million) of convertible bonds with detachable warrants. The five-year bonds have a coupon of between 0.8 and 1.5 percent. The warrants are exercisable during a 10-day period two years after listing. Some products involving securitizations have traded. There have been asset-backed securities, mortgage-backed securities, and the occasional collateralized debt obligation.

Foreign banks have been active in China and have offered complex derivatives involving commodity prices, interest rates, and exchange rates. Unfortunately, some Chinese companies have taken huge losses on these derivatives. For example, Citic Pacific lost \$1.9 billion on derivatives involving the Australian dollar exchange rate and Air China lost \$1.1 billion on oil derivatives. China's regulators took steps to ban speculative derivatives contracts and curtailed the operations of foreign banks in 2009. There has even been speculation that state-owned companies in China will be instructed to default on their derivatives obligations.

36.2 INDIA'S MARKETS

Equity derivatives have a long history in India. Options of various kinds called *teji* (call options), *mandi* (put options), and *fatak* (straddles) traded in unorganized markets as early as 1900 in Mumbai. However, derivatives markets suffered a set back in 1956 when the Securities Control and Regulation Act (SCRA) banned what was considered to be undesirable speculation in securities and again in 1969 when "forward trading contracts" were banned.

The Indian currency is the rupee. In theory, this is freely convertible. In practice, the Reserve Bank of India (RBI: www.rbi.org.in) intervenes to control the exchange rate against the US dollar. (This is sometimes referred to as a dirty or managed float.) In addition, foreign nationals are forbidden from importing or exporting rupees and the extent to which Indian citizens can do this is limited. However, India is moving towards full convertibility.

Securities trading in India is regulated by the Securities and Exchange Board of India (SEBI: www.sebi.gov.in), which was set up in 1988. An important step toward the development of derivatives trading in India was the Securities Laws (Amendment) Ordinance of 1995, which lifted the prohibition on the trading of options. This led the SEBI in 1998 to set up committees to consider how derivatives should be introduced and how risks should be contained. This led to a legal framework, enacted in December 1999, within which derivatives such as options and futures are regarded as securities and can be traded. The trading of stock index futures started in June 2000, and later other products, such as stock index options, stock options, and single stock futures, were allowed.

Since 2000, derivatives trading in India has developed quite fast. Both the National Stock Exchange in Mumbai (NSE: www.nseindia.com) and the Bombay Stock Exchange (BSE: www.bseindia.com) trade index futures, stock futures, index options, and stock options. There are a large number of smaller exchanges trading futures and options. The exchanges use sophisticated technology and are fully electronic. The over-the-counter market is active in India, particularly in interest rate and swap products.

36.3 OTHER DEVELOPING COUNTRIES

Derivatives markets are progressing well in many other developing countries, such as Brazil, Korea, Malaysia, Mexico, Poland, and South Africa. Derivatives are sophisticated instruments and it is necessary for a country's financial markets to reach a reasonable level of development before derivatives can be successful. There are a number of key conditions for the growth of derivatives. It is important for the government to set up a regulatory structure that protects investors from fraud and leaves them feeling sure that their contracts will be honored. However, the government should not impose too many restrictions on the way derivatives can be used because speculators and arbitrageurs are important for the liquidity of the market. There must be a sound financial and legal system within the country. Volatility is not bad for derivatives markets. (Indeed without volatility there would be little interest in many derivative products.) But derivatives are unlikely to thrive unless the economy of the country is reasonably stable and there is a good payments system. Stock markets, bond markets, and money markets should be reasonably well developed. (After all, stocks, bonds, and money market instruments are the underlyings for many derivatives.) Ideally the currency should be freely convertible and there should be no restrictions on the flow of the currency in and out of the country. Moreover, there should be enough swaps, bonds, and money market instruments trading for a risk-free zero-coupon yield curve to be estimated. A final very important condition is that there should be enough well educated individuals who understand the products and how they can be valued.

An intriguing idea for developing countries is the possibility of derivatives transactions between national governments. If country X exports oil to country Y and country Y exports building materials to country X, they are both subject to risks relating to the prices of their exports. It might make sense for them to enter into a swap that effectively fixes prices for several years into the future. This example can be extended so that it applies to groups of countries that trade with each other.¹

SUMMARY

The economies of many developing countries are growing very fast. Indeed, the phrase "developing country" may no longer be an appropriate way of describing them in a decade or two. Although the trading of derivatives occasionally gets out of control

¹ This idea, which has been proposed by Robert Merton, is of course not just appropriate for developing countries.

(see Chapter 35), there can be no question that derivatives markets have played an important role in allowing risks to be managed in developed countries. There is no reason to suppose that the same will not in the future be true in every country of the world. Several developing countries have laid the foundations for a mature derivatives market by creating appropriate legal, financial, and regulatory frameworks. In a few decades, it is likely that the derivatives markets of countries like China, India, and Brazil will be as important as, if not more important than, those of the United States and Western European countries.

FURTHER READING

- Ahuja, N. L., "Commodity Derivatives Market in India: Development, Regulation, and Futures Prospects," *International Research Journal of Finance and Economics*, 2 (2006): 153–62.
- Braga, B. S., "Derivatives Markets in Brazil: An Overview," *Journal of Derivatives*, 4 (Fall 1996): 63–78.
- Ménager-Xu, M. Y., "China's Derivatives Markets," in: *China's Financial Markets: An Insider's Guide to How the Markets Work* (S. N. Neftci and M. Y. Ménager-Xu, eds.). New York: Academic Press, 2007.



Glossary of Terms

ABS *See* Asset-Backed Security.

ABS CDO Instrument where tranches are created from the tranches of ABSs.

Accrual Swap An interest rate swap where interest on one side accrues only when a certain condition is met.

Accrued Interest The interest earned on a bond since the last coupon payment date.

Adaptive Mesh Model A model developed by Figlewski and Gao that grafts a high-resolution tree on to a low-resolution tree so that there is more detailed modeling of the asset price in critical regions.

Agency Costs Costs arising from a situation where the agent (e.g., manager) is not motivated to act in the best interests of the principal (e.g., shareholder).

American Option An option that can be exercised at any time during its life.

Amortizing Swap A swap where the notional principal decreases in a predetermined way as time passes.

Analytic Result Result where answer is in the form of an equation.

Arbitrage A trading strategy that takes advantage of two or more securities being mispriced relative to each other.

Arbitrageur An individual engaging in arbitrage.

Asian Option An option with a payoff dependent on the average price of the underlying asset during a specified period.

Ask Price The price that a dealer is offering to sell an asset.

Asked Price *See* Ask Price.

Asset-Backed Security Security created from a portfolio of loans, bonds, credit card receivables, or other assets.

Asset-or-Nothing Call Option An option that provides a payoff equal to the asset price if the asset price is above the strike price and zero otherwise.

Asset-or-Nothing Put Option An option that provides a payoff equal to the asset price if the asset price is below the strike price and zero otherwise.

Asset Swap Exchanges the coupon on a bond for LIBOR plus a spread.

- As-You-Like-It Option** *See* Chooser Option.
- At-the-Money Option** An option in which the strike price equals the price of the underlying asset.
- Average Price Call Option** An option giving a payoff equal to the greater of zero and the amount by which the average price of the asset exceeds the strike price.
- Average Price Put Option** An option giving a payoff equal to the greater of zero and the amount by which the strike price exceeds the average price of the asset.
- Average Strike Option** An option that provides a payoff dependent on the difference between the final asset price and the average asset price.
- Backdating** Practice (often illegal) of marking a document with a date that precedes the current date.
- Back Testing** Testing a value-at-risk or other model using historical data.
- Backwards Induction** A procedure for working from the end of a tree to its beginning in order to value an option.
- Barrier Option** An option whose payoff depends on whether the path of the underlying asset has reached a barrier (i.e., a certain predetermined level).
- Base Correlation** Correlation that leads to the price of a 0% to X% CDO tranche being consistent with the market for a particular value of X.
- Basel Committee** Committee responsible for regulation of banks internationally.
- Basis** The difference between the spot price and the futures price of a commodity.
- Basis Point** When used to describe an interest rate, a basis point is one hundredth of one percent (= 0.01%)
- Basis Risk** The risk to a hedger arising from uncertainty about the basis at a future time.
- Basis Swap** A swap where cash flows determined by one floating reference rate are exchanged for cash flows determined by another floating reference rate.
- Basket Credit Default Swap** Credit default swap where there are several reference entities.
- Basket Option** An option that provides a payoff dependent on the value of a portfolio of assets.
- Bear Spread** A short position in a put option with strike price K_1 combined with a long position in a put option with strike price K_2 where $K_2 > K_1$. (A bear spread can also be created with call options.)
- Bermudan Option** An option that can be exercised on specified dates during its life.
- Beta** A measure of the systematic risk of an asset.
- Bid-Ask Spread** The amount by which the ask price exceeds the bid price.
- Bid-Offer Spread** *See* Bid-Ask Spread.
- Bid Price** The price that a dealer is prepared to pay for an asset.
- Binary Credit Default Swap** Instrument where there is a fixed dollar payoff in the event of a default by a particular company.
- Binary Option** Option with a discontinuous payoff, e.g., a cash-or-nothing option or an asset-or-nothing option.

- Binomial Model** A model where the price of an asset is monitored over successive short periods of time. In each short period it is assumed that only two price movements are possible.
- Binomial Tree** A tree that represents how an asset price can evolve under the binomial model.
- Bivariate Normal Distribution** A distribution for two correlated variables, each of which is normal.
- Black's Approximation** An approximate procedure developed by Fischer Black for valuing a call option on a dividend-paying stock.
- Black's Model** An extension of the Black-Scholes model for valuing European options on futures contracts. As described in Chapter 26, it is used extensively in practice to value European options when the distribution of the asset price at maturity is assumed to be lognormal.
- Black-Scholes-Merton Model** A model for pricing European options on stocks, developed by Fischer Black, Myron Scholes, and Robert Merton.
- Bond Option** An option where a bond is the underlying asset.
- Bond Yield** Discount rate which, when applied to all the cash flows of a bond, causes the present value of the cash flows to equal the bond's market price.
- Bootstrap Method** A procedure for calculating the zero-coupon yield curve from market data.
- Boston Option** *See* Deferred Payment Option.
- Box Spread** A combination of a bull spread created from calls and a bear spread created from puts.
- Break Forward** *See* Deferred Payment Option.
- Brownian Motion** *See* Wiener Process.
- Bull Spread** A long position in a call with strike price K_1 combined with a short position in a call with strike price K_2 , where $K_2 > K_1$. (A bull spread can also be created with put options.)
- Butterfly Spread** A position that is created by taking a long position in a call with strike price K_1 , a long position in a call with strike price K_3 , and a short position in two calls with strike price K_2 , where $K_3 > K_2 > K_1$ and $K_2 = 0.5(K_1 + K_3)$. (A butterfly spread can also be created with put options.)
- Calendar Spread** A position that is created by taking a long position in a call option that matures at one time and a short position in a similar call option that matures at a different time. (A calendar spread can also be created using put options.)
- Calibration** Method for implying a model's parameters from the prices of actively traded options.
- Callable Bond** A bond containing provisions that allow the issuer to buy it back at a predetermined price at certain times during its life.
- Call Option** An option to buy an asset at a certain price by a certain date.
- Cancelable Swap** Swap that can be canceled by one side on prespecified dates.
- Cap** *See* Interest Rate Cap.

- Cap Rate** The rate determining payoffs in an interest rate cap.
- Capital Asset Pricing Model** A model relating the expected return on an asset to its beta.
- Caplet** One component of an interest rate cap.
- Case-Shiller Index** Index of house prices in the United States.
- Cash Flow Mapping** A procedure for representing an instrument as a portfolio of zero-coupon bonds for the purpose of calculating value at risk.
- Cash-or-Nothing Call Option** An option that provides a fixed predetermined payoff if the final asset price is above the strike price and zero otherwise.
- Cash-or-Nothing Put Option** An option that provides a fixed predetermined payoff if the final asset price is below the strike price and zero otherwise.
- Cash Settlement** Procedure for settling a futures contract in cash rather than by delivering the underlying asset.
- CAT Bond** Bond where the interest and, possibly, the principal paid are reduced if a particular category of "catastrophic" insurance claims exceed a certain amount.
- CDD** Cooling degree days. The maximum of zero and the amount by which the daily average temperature is greater than 65° Fahrenheit. The average temperature is the average of the highest and lowest temperatures (midnight to midnight).
- CDO** *See* Collateralized Debt Obligation.
- CDO Squared** An instrument in which the default risks in a portfolio of CDO tranches are allocated to new securities.
- CDX NA IG** Portfolio of 125 North American companies.
- Central Clearing Party** A clearing house used for over-the-counter contracts.
- Cheapest-to-Deliver Bond** The bond that is cheapest to deliver in the Chicago Board of Trade bond futures contract.
- Cholesky Decomposition** A method of sampling from a multivariate normal distribution.
- Chooser Option** An option where the holder has the right to choose whether it is a call or a put at some point during its life.
- Class of Options** *See* Option Class.
- Clean Price of Bond** The quoted price of a bond. The cash price paid for the bond (or dirty price) is calculated by adding the accrued interest to the clean price.
- Clearing House** A firm that guarantees the performance of the parties in a derivatives transaction (also referred to as a clearing corporation).
- Clearing Margin** A margin posted by a member of a clearinghouse.
- Cliquet Option** A series of call or put options with rules for determining strike prices. Typically, one option starts when the previous one terminates.
- CMO** Collateralized Mortgage Obligation.
- Collar** *See* Interest Rate Collar.
- Collateralization** A system for posting collateral by one or both parties in a derivatives transaction.

- Collateralized Debt Obligation** A way of packaging credit risk. Several classes of securities (known as tranches) are created from a portfolio of bonds and there are rules for determining how the cost of defaults are allocated to classes.
- Collateralized Mortgage Obligation (CMO)** A mortgage-backed security where investors are divided into classes and there are rules for determining how principal repayments are channeled to the classes.
- Combination** A position involving both calls and puts on the same underlying asset.
- Commodity Futures Trading Commission** A body that regulates trading in futures contracts in the United States.
- Commodity Swap** A swap where cash flows depend on the price of a commodity.
- Compound Correlation** Correlation implied from the market price of a CDO tranche.
- Compound Option** An option on an option.
- Compounding Frequency** This defines how an interest rate is measured.
- Compounding Swap** Swap where interest compounds instead of being paid.
- Conditional Value at Risk (C-VaR)** Expected loss during N days conditional on being in the $(100 - X)\%$ tail of the distribution of profits/losses. The variable N is the time horizon and $X\%$ is the confidence level.
- Confirmation** Contract confirming verbal agreement between two parties to a trade in the over-the-counter market.
- Constant Elasticity of Variance (CEV) Model** Model where the variance of the change in a variable in a short period of time is proportional to the value of the variable.
- Constant Maturity Swap (CMS)** A swap where a swap rate is exchanged for either a fixed rate or a floating rate on each payment date.
- Constant Maturity Treasury Swap** A swap where the yield on a Treasury bond is exchanged for either a fixed rate or a floating rate on each payment date.
- Consumption Asset** An asset held for consumption rather than investment.
- Contango** A situation where the futures price is above the expected future spot price (also often used to refer to the situation where the futures price is above the current spot price).
- Continuous Compounding** A way of quoting interest rates. It is the limit as the assumed compounding interval is made smaller and smaller.
- Control Variate Technique** A technique that can sometimes be used for improving the accuracy of a numerical procedure.
- Convenience Yield** A measure of the benefits from ownership of an asset that are not obtained by the holder of a long futures contract on the asset.
- Conversion Factor** A factor used to determine the number of bonds that must be delivered in the Chicago Board of Trade bond futures contract.
- Convertible Bond** A corporate bond that can be converted into a predetermined amount of the company's equity at certain times during its life.
- Convexity** A measure of the curvature in the relationship between bond prices and bond yields.

- Convexity Adjustment** An overworked term. For example, it can refer to the adjustment necessary to convert a futures interest rate to a forward interest rate. It can also refer to the adjustment to a forward rate that is sometimes necessary when Black's model is used.
- Copula** A way of defining the correlation between variables with known distributions.
- Cornish–Fisher Expansion** An approximate relationship between the fractiles of a probability distribution and its moments.
- Cost of Carry** The storage costs plus the cost of financing an asset minus the income earned on the asset.
- Counterparty** The opposite side in a financial transaction.
- Coupon** Interest payment made on a bond.
- Covariance** Measure of the linear relationship between two variables (equals the correlation between the variables times the product of their standard deviations).
- Covariance Matrix** See Variance–Covariance Matrix.
- Covered Call** A short position in a call option on an asset combined with a long position in the asset.
- Crashophobia** Fear of a stock market crash that some people claim causes the market to increase the price of deep-out-of-the-money put options.
- Credit Contagion** The tendency of a default by one company to lead to defaults by other companies.
- Credit Default Swap** An instrument that gives the holder the right to sell a bond for its face value in the event of a default by the issuer.
- Credit Derivative** A derivative whose payoff depends on the creditworthiness of one or more companies or countries.
- Credit Index** Index that tracks the cost of buying protection for each company in a portfolio (e.g., CDX NA IG and iTraxx Europe).
- Credit Rating** A measure of the creditworthiness of a bond issue.
- Credit Ratings Transition Matrix** A table showing the probability that a company will move from one credit rating to another during a certain period of time.
- Credit Risk** The risk that a loss will be experienced because of a default by the counterparty in a derivatives transaction.
- Credit Spread Option** Option whose payoff depends on the spread between the yields earned on two assets.
- Credit Value Adjustment** Adjustment to value of derivatives outstanding with a counterparty to reflect the counterparty's default risk.
- Credit Value at Risk** The credit loss that will not be exceeded at some specified confidence level.
- CreditMetrics** A procedure for calculating credit value at risk.
- Cross Hedging** Hedging an exposure to the price of one asset with a contract on another asset.
- Cumulative Distribution Function** The probability that a variable will be less than x as a function of x .

Currency Swap A swap where interest and principal in one currency are exchanged for interest and principal in another currency.

CVA *See* Credit Value Adjustment.

Day Count A convention for quoting interest rates.

Day Trade A trade that is entered into and closed out on the same day.

Default Correlation Measures the tendency of two companies to default at about the same time.

Default Intensity *See* Hazard Rate.

Default Probability Density Measures the unconditional probability of default in a future short period of time.

Deferred Payment Option An option where the price paid is deferred until the end of the option's life.

Deferred Swap An agreement to enter into a swap at some time in the future (also called a forward swap).

Delivery Price Price agreed to (possibly some time in the past) in a forward contract.

Delta The rate of change of the price of a derivative with the price of the underlying asset.

Delta Hedging A hedging scheme that is designed to make the price of a portfolio of derivatives insensitive to small changes in the price of the underlying asset.

Delta-Neutral Portfolio A portfolio with a delta of zero so that there is no sensitivity to small changes in the price of the underlying asset.

DerivaGem The software accompanying this book.

Derivative An instrument whose price depends on, or is derived from, the price of another asset.

Deterministic Variable A variable whose future value is known.

Diagonal Spread A position in two calls where both the strike prices and times to maturity are different. (A diagonal spread can also be created with put options.)

Differential Swap A swap where a floating rate in one currency is exchanged for a floating rate in another currency and both rates are applied to the same principal.

Diffusion Process Model where value of asset changes continuously (no jumps).

Dirty Price of Bond Cash price of bond.

Discount Bond *See* Zero-Coupon Bond.

Discount Instrument An instrument, such as a Treasury bill, that provides no coupons.

Diversification Reducing risk by dividing a portfolio between many different assets.

Dividend A cash payment made to the owner of a stock.

Dividend Yield The dividend as a percentage of the stock price.

Dollar Duration The product of a bond's modified duration and the bond price.

Down-and-In Option An option that comes into existence when the price of the underlying asset declines to a prespecified level.

- Down-and-Out Option** An option that ceases to exist when the price of the underlying asset declines to a prespecified level.
- Downgrade Trigger** A clause in a contract that states that the contract will be terminated with a cash settlement if the credit rating of one side falls below a certain level.
- Drift Rate** The average increase per unit of time in a stochastic variable.
- Duration** A measure of the average life a bond. It is also an approximation to the ratio of the proportional change in the bond price to the absolute change in its yield.
- Duration Matching** A procedure for matching the durations of assets and liabilities in a financial institution.
- DV01** The dollar value of a 1-basis-point increase in all interest rates.
- Dynamic Hedging** A procedure for hedging an option position by periodically changing the position held in the underlying asset. The objective is usually to maintain a delta-neutral position.
- Early Exercise** Exercise prior to the maturity date.
- Efficient Market Hypothesis** A hypothesis that asset prices reflect relevant information.
- Electronic Trading** System of trading where a computer is used to match buyers and sellers.
- Embedded Option** An option that is an inseparable part of another instrument.
- Empirical Research** Research based on historical market data.
- Employee Stock Option** A stock option issued by company on its own stock and given to its employees as part of their remuneration.
- Equilibrium Model** A model for the behavior of interest rates derived from a model of the economy.
- Equity Swap** A swap where the return on an equity portfolio is exchanged for either a fixed or a floating rate of interest.
- Equity Tranche** The tranche that first absorbs losses.
- Eurocurrency** A currency that is outside the formal control of the issuing country's monetary authorities.
- Eurodollar** A dollar held in a bank outside the United States.
- Eurodollar Futures Contract** A futures contract written on a Eurodollar deposit.
- Eurodollar Interest Rate** The interest rate on a Eurodollar deposit.
- European Option** An option that can be exercised only at the end of its life.
- EWMA** Exponentially weighted moving average.
- Exchange Option** An option to exchange one asset for another.
- Ex-dividend Date** When a dividend is declared, an ex-dividend date is specified. Investors who own shares of the stock just before the ex-dividend date receive the dividend.
- Exercise Limit** Maximum number of option contracts that can be exercised within a five-day period.

- Exercise Multiple** Ratio of stock price to strike price at time of exercise for employee stock option.
- Exercise Price** The price at which the underlying asset may be bought or sold in an option contract (also called the strike price).
- Exotic Option** A nonstandard option.
- Expectations Theory** The theory that forward interest rates equal expected future spot interest rates.
- Expected Shortfall** See Conditional Value at Risk.
- Expected Value of a Variable** The average value of the variable obtained by weighting the alternative values by their probabilities.
- Expiration Date** The end of life of a contract.
- Explicit Finite Difference Method** A method for valuing a derivative by solving the underlying differential equation. The value of the derivative at time t is related to three values at time $t + \Delta t$. It is essentially the same as the trinomial tree method.
- Exponentially Weighted Moving Average Model** A model where exponential weighting is used to provide forecasts for a variable from historical data. It is sometimes applied to variances and covariances in value at risk calculations.
- Exponential Weighting** A weighting scheme where the weight given to an observation depends on how recent it is. The weight given to an observation i time periods ago is λ times the weight given to an observation $i - 1$ time periods ago where $\lambda < 1$.
- Exposure** The maximum loss from default by a counterparty.
- Extendable Bond** A bond whose life can be extended at the option of the holder.
- Extendable Swap** A swap whose life can be extended at the option of one side to the contract.
- Factor** Source of uncertainty.
- Factor analysis** An analysis aimed at finding a small number of factors that describe most of the variation in a large number of correlated variables (similar to a principal components analysis).
- FAS 123** Accounting standard in United States relating to employee stock options.
- FAS 133** Accounting standard in United States relating to instruments used for hedging.
- FASB** Financial Accounting Standards Board.
- FICO** A credit score developed by Fair Isaac Corporation.
- Financial Intermediary** A bank or other financial institution that facilitates the flow of funds between different entities in the economy.
- Finite Difference Method** A method for solving a differential equation.
- Flat Volatility** The name given to volatility used to price a cap when the same volatility is used for each caplet.
- Flex Option** An option traded on an exchange with terms that are different from the standard options traded by the exchange.
- Flexi Cap** Interest rate cap where there is a limit on the total number of caplets that can be exercised.

Floor *See* Interest Rate Floor.

Floor–Ceiling Agreement *See* Collar.

Floorlet One component of a floor.

Floor Rate The rate in an interest rate floor agreement.

Foreign Currency Option An option on a foreign exchange rate.

Forward Contract A contract that obligates the holder to buy or sell an asset for a predetermined delivery price at a predetermined future time.

Forward Exchange Rate The forward price of one unit of a foreign currency.

Forward Interest Rate The interest rate for a future period of time implied by the rates prevailing in the market today.

Forward Price The delivery price in a forward contract that causes the contract to be worth zero.

Forward Rate Rate of interest for a period of time in the future implied by today's zero rates.

Forward Rate Agreement (FRA) Agreement that a certain interest rate will apply to a certain principal amount for a certain time period in the future.

Forward Risk-Neutral World A world is forward risk-neutral with respect to a certain asset when the market price of risk equals the volatility of that asset.

Forward Start Option An option designed so that it will be at-the-money at some time in the future.

Forward Swap *See* Deferred Swap.

Futures Commission Merchants Futures traders who are following instructions from clients.

Futures Contract A contract that obligates the holder to buy or sell an asset at a predetermined delivery price during a specified future time period. The contract is settled daily.

Futures Option An option on a futures contract.

Futures Price The delivery price currently applicable to a futures contract.

Futures-Style Option Futures contract on the payoff from an option. /

Gamma The rate of change of delta with respect to the asset price.

Gamma-Neutral Portfolio A portfolio with a gamma of zero.

Gap Option European call or put option where there are two strike prices. One determines whether the option is exercised. The other determines the payoff.

GARCH Model A model for forecasting volatility where the variance rate follows a mean-reverting process.

Gaussian Copula Model A model for defining a correlation structure between two or more variables. In some credit derivatives models, it is used to define a correlation structure for times to default.

Gaussian Quadrature Procedure for integrating over a normal distribution.

Generalized Wiener Process A stochastic process where the change in a variable in time t has a normal distribution with mean and variance both proportional to t .

- Geometric Average** The n th root of the product of n numbers.
- Geometric Brownian Motion** A stochastic process often assumed for asset prices where the logarithm of the underlying variable follows a generalized Wiener process.
- Girsanov's Theorem** Result showing that when we change the measure (e.g., move from real world to risk-neutral world) the expected return of a variable changes but the volatility remains the same.
- Greeks** Hedge parameters such as delta, gamma, vega, theta, and rho.
- Haircut** Discount applied to the value of an asset for collateral purposes.
- Hazard Rate** Measures probability of default in a short period of time conditional on no earlier default.
- HDD** Heating degree days. The maximum of zero and the amount by which the daily average temperature is less than 65° Fahrenheit. The average temperature is the average of the highest and lowest temperatures (midnight to midnight).
- Hedge** A trade designed to reduce risk.
- Hedge Funds** Funds that are subject to less regulation and fewer restrictions than mutual funds. They can take short positions and use derivatives, but they cannot publicly offer their securities.
- Hedger** An individual who enters into hedging trades.
- Hedge Ratio** The ratio of the size of a position in a hedging instrument to the size of the position being hedged.
- Historical Simulation** A simulation based on historical data.
- Historic Volatility** A volatility estimated from historical data.
- Holiday Calendar** Calendar defining which days are holidays for the purposes of determining payment dates in a swap.
- IMM Dates** Third Wednesday in March, June, September, and December.
- Implicit Finite Difference Method** A method for valuing a derivative by solving the underlying differential equation. The value of the derivative at time $t + \Delta t$ is related to three values at time t .
- Implied Correlation** Correlation number implied from the price of a credit derivative using the Gaussian copula or similar model.
- Implied Distribution** A distribution for a future asset price implied from option prices.
- Implied Tree** A tree describing the movements of an asset price that is constructed to be consistent with observed option prices.
- Implied Volatility** Volatility implied from an option price using the Black-Scholes or a similar model.
- Implied Volatility Function (IVF) Model** Model designed so that it matches the market prices of all European options.
- Inception Profit** Profit created by selling a derivative for more than its theoretical value.
- Index Amortizing Swap** See indexed principal swap.

- Index Arbitrage** An arbitrage involving a position in the stocks comprising a stock index and a position in a futures contract on the stock index.
- Index Futures** A futures contract on a stock index or other index.
- Index Option** An option contract on a stock index or other index.
- Indexed Principal Swap** A swap where the principal declines over time. The reduction in the principal on a payment date depends on the level of interest rates.
- Initial Margin** The cash required from a futures trader at the time of the trade.
- Instantaneous Forward Rate** Forward rate for a very short period of time in the future.
- Interest Rate Cap** An option that provides a payoff when a specified interest rate is above a certain level. The interest rate is a floating rate that is reset periodically.
- Interest Rate Collar** A combination of an interest-rate cap and an interest rate floor.
- Interest Rate Derivative** A derivative whose payoffs are dependent on future interest rates.
- Interest Rate Floor** An option that provides a payoff when an interest rate is below a certain level. The interest rate is a floating rate that is reset periodically.
- Interest Rate Option** An option where the payoff is dependent on the level of interest rates.
- Interest Rate Swap** An exchange of a fixed rate of interest on a certain notional principal for a floating rate of interest on the same notional principal.
- International Swaps and Derivatives Association** Trade Association for over-the-counter derivatives and developer of master agreements used in over-the-counter contracts.
- In-the-Money Option** Either (a) a call option where the asset price is greater than the strike price or (b) a put option where the asset price is less than the strike price.
- Intrinsic Value** For a call option, this is the greater of the excess of the asset price over the strike price and zero. For a put option, it is the greater of the excess of the strike price over the asset price and zero.
- Inverted Market** A market where futures prices decrease with maturity.
- Investment Asset** An asset held by at least some individuals for investment purposes.
- IO** Interest Only. A mortgage-backed security where the holder receives only interest cash flows on the underlying mortgage pool.
- ISDA** See International Swaps and Derivatives Association.
- Itô Process** A stochastic process where the change in a variable during each short period of time of length Δt has a normal distribution. The mean and variance of the distribution are proportional to Δt and are not necessarily constant.
- Itô's Lemma** A result that enables the stochastic process for a function of a variable to be calculated from the stochastic process for the variable itself.
- ITraxx Europe** Portfolio of 125 investment-grade European companies.
- Jump-Diffusion Model** Model where asset price has jumps superimposed on to a diffusion process such as geometric Brownian motion.

- Jump Process** Stochastic process for a variable involving jumps in the value of the variable.
- Kurtosis** A measure of the fatness of the tails of a distribution.
- LEAPS** Long-term equity anticipation securities. These are relatively long-term options on individual stocks or stock indices.
- LIBID** London interbank bid rate. The rate bid by banks on Eurocurrency deposits (i.e., the rate at which a bank is willing to borrow from other banks).
- LIBOR** London interbank offered rate. The rate offered by banks on Eurocurrency deposits (i.e., the rate at which a bank is willing to lend to other banks).
- LIBOR Curve** LIBOR zero-coupon interest rates as a function of maturity.
- LIBOR-in-Arrears Swap** Swap where the interest paid on a date is determined by the interest rate observed on that date (not by the interest rate observed on the previous payment date).
- Limit Move** The maximum price move permitted by the exchange in a single trading session.
- Limit Order** An order that can be executed only at a specified price or one more favorable to the investor.
- Liquidity Preference Theory** A theory leading to the conclusion that forward interest rates are above expected future spot interest rates.
- Liquidity Premium** The amount that forward interest rates exceed expected future spot interest rates.
- Liquidity Risk** Risk that it will not be possible to sell a holding of a particular instrument at its theoretical price. Also, the risk that a company will not be able to borrow money to fund its assets.
- Locals** Individuals on the floor of an exchange who trade for their own account rather than for someone else.
- Lognormal Distribution** A variable has a lognormal distribution when the logarithm of the variable has a normal distribution.
- Long Hedge** A hedge involving a long futures position.
- Long Position** A position involving the purchase of an asset.
- Lookback Option** An option whose payoff is dependent on the maximum or minimum of the asset price achieved during a certain period.
- Low Discrepancy Sequence** See Quasi-random Sequence.
- Maintenance Margin** When the balance in a trader's margin account falls below the maintenance margin level, the trader receives a margin call requiring the account to be topped up to the initial margin level.
- Margin** The cash balance (or security deposit) required from a futures or options trader.
- Margin Call** A request for extra margin when the balance in the margin account falls below the maintenance margin level.
- Market Maker** A trader who is willing to quote both bid and offer prices for an asset.
- Market Model** A model most commonly used by traders.

- Market Price of Risk** A measure of the trade-offs investors make between risk and return.
- Market Segmentation Theory** A theory that short interest rates are determined independently of long interest rates by the market.
- Marking to Market** The practice of revaluing an instrument to reflect the current values of the relevant market variables.
- Markov Process** A stochastic process where the behavior of the variable over a short period of time depends solely on the value of the variable at the beginning of the period, not on its past history.
- Martingale** A zero drift stochastic process.
- Maturity Date** The end of the life of a contract.
- Maximum Likelihood Method** A method for choosing the values of parameters by maximizing the probability of a set of observations occurring.
- Mean Reversion** The tendency of a market variable (such as an interest rate) to revert back to some long-run average level.
- Measure** Sometimes also called a probability measure, it defines the market price of risk.
- Mezzanine Tranche** Tranche which experiences losses after equity tranche but before senior tranches.
- Modified Duration** A modification to the standard duration measure so that it more accurately describes the relationship between proportional changes in a bond price and actual changes in its yield. The modification takes account of the compounding frequency with which the yield is quoted.
- Money Market Account** An investment that is initially equal to \$1 and, at time t , increases at the very short-term risk-free interest rate prevailing at that time.
- Monte Carlo Simulation** A procedure for randomly sampling changes in market variables in order to value a derivative.
- Mortgage-Backed Security** A security that entitles the owner to a share in the cash flows realized from a pool of mortgages.
- Naked Position** A short position in a call option that is not combined with a long position in the underlying asset.
- Netting** The ability to offset contracts with positive and negative values in the event of a default by a counterparty.
- Newton-Raphson Method** An iterative procedure for solving nonlinear equations.
- NINJA** Term used to describe a person with a poor credit risk: no income, no job, no assets.
- No-Arbitrage Assumption** The assumption that there are no arbitrage opportunities in market prices.
- No-Arbitrage Interest Rate Model** A model for the behavior of interest rates that is exactly consistent with the initial term structure of interest rates.
- Nonstationary Model** A model where the volatility parameters are a function of time.
- Nonsystematic Risk** Risk that can be diversified away.

Normal Backwardation A situation where the futures price is below the expected future spot price.

Normal Distribution The standard bell-shaped distribution of statistics.

Normal Market A market where futures prices increase with maturity.

Notional Principal The principal used to calculate payments in an interest rate swap. The principal is "notional" because it is neither paid nor received.

Numeraire Defines the units in which security prices are measured. For example, if the price of IBM is the numeraire, all security prices are measured relative to IBM. If IBM is \$80 and a particular security price is \$50, the security price is 0.625 when IBM is the numeraire.

Numerical Procedure A method of valuing an option when no formula is available.

OCC Options Clearing Corporation. *See* Clearinghouse.

Offer Price *See* Ask Price.

OIS *See* Overnight Indexed Swap.

Open Interest The total number of long positions outstanding in a futures contract (equals the total number of short positions).

Open Outcry System of trading where traders meet on the floor of the exchange

Option The right to buy or sell an asset.

Option-Adjusted Spread The spread over the Treasury curve that makes the theoretical price of an interest rate derivative equal to the market price.

Option Class All options of the same type (call or put) on a particular stock.

Option Series All options of a certain class with the same strike price and expiration date.

Order Book Official *See* Board Broker.

Out-of-the-Money Option Either (a) a call option where the asset price is less than the strike price or (b) a put option where the asset price is greater than the strike price.

Overnight Indexed Swap Swap where a fixed rate for a period (e.g., 1 month) is exchanged for the geometric average of the overnight rates during the period.

Over-the-Counter Market A market where traders deal by phone. The traders are usually financial institutions, corporations, and fund managers.

Package A derivative that is a portfolio of standard calls and puts, possibly combined with a position in forward contracts and the asset itself.

Par Value The principal amount of a bond.

Par Yield The coupon on a bond that makes its price equal the principal.

Parallel Shift A movement in the yield curve where each point on the curve changes by the same amount.

Parisian Option Barrier option where the asset has to be above or below the barrier for a period of time for the option to be knocked in or out.

Path-Dependent Option An option whose payoff depends on the whole path followed by the underlying variable—not just its final value.

- Payoff** The cash realized by the holder of an option or other derivative at the end of its life.
- PD** Probability of default.
- Plain Vanilla** A term used to describe a standard deal.
- P-Measure** Real-world measure.
- PO** Principal Only. A mortgage-backed security where the holder receives only principal cash flows on the underlying mortgage pool.
- Poisson Process** A process describing a situation where events happen at random. The probability of an event in time Δt is $\lambda \Delta t$, where λ is the intensity of the process.
- Portfolio Immunization** Making a portfolio relatively insensitive to interest rates.
- Portfolio Insurance** Entering into trades to ensure that the value of a portfolio will not fall below a certain level.
- Position Limit** The maximum position a trader (or group of traders acting together) is allowed to hold.
- Premium** The price of an option.
- Prepayment function** A function estimating the prepayment of principal on a portfolio of mortgages in terms of other variables.
- Principal** The par or face value of a debt instrument.
- Principal Components Analysis** An analysis aimed at finding a small number of factors that describe most of the variation in a large number of correlated variables (similar to a factor analysis).
- Principal Protected Note** A product where the return earned depends on the performance of a risky asset but is guaranteed to be nonnegative, so that the investor's principal is preserved.
- Program Trading** A procedure where trades are automatically generated by a computer and transmitted to the trading floor of an exchange.
- Protective Put** A put option combined with a long position in the underlying asset.
- Pull-to-Par** The reversion of a bond's price to its par value at maturity.
- Put-Call Parity** The relationship between the price of a European call option and the price of a European put option when they have the same strike price and maturity date.
- Put Option** An option to sell an asset for a certain price by a certain date.
- Puttable Bond** A bond where the holder has the right to sell it back to the issuer at certain predetermined times for a predetermined price.
- Puttable Swap** A swap where one side has the right to terminate early.
- Q-Measure** Risk-neutral measure.
- Quanto** A derivative where the payoff is defined by variables associated with one currency but is paid in another currency.
- Quasi-random Sequences** A sequences of numbers used in a Monte Carlo simulation that are representative of alternative outcomes rather than random.
- Rainbow Option** An option whose payoff is dependent on two or more underlying variables.

- Range Forward Contract** The combination of a long call and short put or the combination of a short call and long put.
- Ratchet Cap** Interest rate cap where the cap rate applicable to an accrual period equals the rate for the previous accrual period plus a spread.
- Real Option** Option involving real (as opposed to financial) assets. Real assets include land, plant, and machinery.
- Rebalancing** The process of adjusting a trading position periodically. Usually the purpose is to maintain delta neutrality.
- Recovery Rate** Amount recovered in the event of a default as a percent of the face value.
- Reference Entity** Company for which default protection is bought in a credit default swap.
- Repo** Repurchase agreement. A procedure for borrowing money by selling securities to a counterparty and agreeing to buy them back later at a slightly higher price.
- Repo Rate** The rate of interest in a repo transaction.
- Reset Date** The date in a swap or cap or floor when the floating rate for the next period is set.
- Reversion Level** The level to which the value of a market variable (e.g., an interest rate) tends to revert.
- Rho** Rate of change of the price of a derivative with the interest rate.
- Rights Issue** An issue to existing shareholders of a security giving them the right to buy new shares at a certain price.
- Risk-Free Rate** The rate of interest that can be earned without assuming any risks.
- Risk-Neutral Valuation** The valuation of an option or other derivative assuming the world is risk neutral. Risk-neutral valuation gives the correct price for a derivative in all worlds, not just in a risk-neutral world.
- Risk-Neutral World** A world where investors are assumed to require no extra return on average for bearing risks.
- Roll Back** See Backwards Induction.
- Scalper** A trader who holds positions for a very short period of time.
- Scenario Analysis** An analysis of the effects of possible alternative future movements in market variables on the value of a portfolio.
- SEC** Securities and Exchange Commission.
- Securitization** Procedure for distributing the risks in a portfolio of assets.
- Settlement Price** The average of the prices that a contract trades for immediately before the bell signaling the close of trading for a day. It is used in mark-to-market calculations.
- Sharpe Ratio** Ratio of excess return over risk-free rate to standard deviation of the excess return.
- Short Hedge** A hedge where a short futures position is taken.
- Short Position** A position assumed when traders sell shares they do not own.
- Short Rate** The interest rate applying for a very short period of time.

Short Selling Selling in the market shares that have been borrowed from another investor.

Short-Term Risk-Free Rate *See* Short Rate.

Shout Option An option where the holder has the right to lock in a minimum value for the payoff at one time during its life.

Simulation *See* Monte Carlo Simulation.

Specialist An individual responsible for managing limit orders on some exchanges. The specialist does not make the information on outstanding limit orders available to other traders.

Speculator An individual who is taking a position in the market. Usually the individual is betting that the price of an asset will go up or that the price of an asset will go down.

Spot Interest Rate *See* Zero-Coupon Interest Rate.

Spot Price The price for immediate delivery.

Spot Volatilities The volatilities used to price a cap when a different volatility is used for each caplet.

Spread Option An option where the payoff is dependent on the difference between two market variables.

Spread Transaction A position in two or more options of the same type.

Stack and Roll Procedure where short-term futures contracts are rolled forward so that long-term hedges are created.

Static Hedge A hedge that does not have to be changed once it is initiated.

Static Options Replication A procedure for hedging a portfolio that involves finding another portfolio of approximately equal value on some boundary.

Step-up Swap A swap where the principal increases over time in a predetermined way.

Sticky Cap Interest rate cap where the cap rate applicable to an accrual period equals the capped rate for the previous accrual period plus a spread.

Stochastic Process An equation describing the probabilistic behavior of a stochastic variable.

Stochastic Variable A variable whose future value is uncertain.

Stock Dividend A dividend paid in the form of additional shares.

Stock Index An index monitoring the value of a portfolio of stocks.

Stock Index Futures Futures on a stock index.

Stock Index Option An option on a stock index.

Stock Option Option on a stock.

Stock Split The conversion of each existing share into more than one new share.

Storage Costs The costs of storing a commodity.

Straddle A long position in a call and a put with the same strike price.

Strangle A long position in a call and a put with different strike prices.

Strap A long position in two call options and one put option with the same strike price.

- Stressed VaR** Value at risk calculated using historical simulation from a period of stressed market conditions.
- Stress Testing** Testing of the impact of extreme market moves on the value of a portfolio.
- Strike Price** The price at which the asset may be bought or sold in an option contract (also called the exercise price).
- Strip** A long position in one call option and two put options with the same strike price.
- Strip Bonds** Zero-coupon bonds created by selling the coupons on Treasury bonds separately from the principal.
- Subprime Mortgage** Mortgage granted to borrower with a poor credit history or no credit history.
- Swap** An agreement to exchange cash flows in the future according to a prearranged formula.
- Swap Rate** The fixed rate in an interest rate swap that causes the swap to have a value of zero.
- Swaption** An option to enter into an interest rate swap where a specified fixed rate is exchanged for floating.
- Swing Option** Energy option in which the rate of consumption must be between a minimum and maximum level. There is usually a limit on the number of times the option holder can change the rate at which the energy is consumed.
- Synthetic CDO** A CDO created by selling credit default swaps.
- Synthetic Option** An option created by trading the underlying asset.
- Systematic Risk** Risk that cannot be diversified away.
- Systemic Risk** Risk that a default by one financial institution will lead to defaults by other financial institutions.
- Tailing the Hedge** A procedure for adjusting the number of futures contracts used for hedging to reflect daily settlement.
- Tail Loss** *See* Conditional Value at Risk.
- Take-and-Pay Option** *See* Swing Option.
- TED Spread** The difference between 3-month LIBOR and the 3-month T-Bill rate.
- Term Structure of Interest Rates** The relationship between interest rates and their maturities.
- Terminal Value** The value at maturity.
- Theta** The rate of change of the price of an option or other derivative with the passage of time.
- Time Decay** *See* Theta.
- Time Value** The value of an option arising from the time left to maturity (equals an option's price minus its intrinsic value).
- Timing Adjustment** Adjustment made to the forward value of a variable to allow for the timing of a payoff from a derivative.

- Total Return Swap** A swap where the return on an asset such as a bond is exchanged for LIBOR plus a spread. The return on the asset includes income such as coupons and the change in value of the asset.
- Tranche** One of several securities that have different risk attributes. Examples are the tranches of a CDO or CMO.
- Transaction Costs** The cost of carrying out a trade (commissions plus the difference between the price obtained and the midpoint of the bid-offer spread).
- Treasury Bill** A short-term non-coupon-bearing instrument issued by the government to finance its debt.
- Treasury Bond** A long-term coupon-bearing instrument issued by the government to finance its debt.
- Treasury Bond Futures** A futures contract on Treasury bonds.
- Treasury Note** *See* Treasury Bond. (Treasury notes have maturities of less than 10 years.)
- Treasury Note Futures** A futures contract on Treasury notes.
- Tree** Representation of the evolution of the value of a market variable for the purposes of valuing an option or other derivative.
- Trinomial Tree** A tree where there are three branches emanating from each node. It is used in the same way as a binomial tree for valuing derivatives.
- Triple Witching Hour** A term given to the time when stock index futures, stock index options, and options on stock index futures all expire together.
- Underlying Variable** A variable on which the price of an option or other derivative depends.
- Unsystematic Risk** *See* Nonsystematic Risk.
- Up-and-In Option** An option that comes into existence when the price of the underlying asset increases to a prespecified level.
- Up-and-Out Option** An option that ceases to exist when the price of the underlying asset increases to a prespecified level.
- Uptick** An increase in price.
- Value at Risk** A loss that will not be exceeded at some specified confidence level.
- Variance-Covariance Matrix** A matrix showing variances of, and covariances between, a number of different market variables.
- Variance-Gamma Model** A pure jump model where small jumps occur often and large jumps occur infrequently.
- Variance Rate** The square of volatility.
- Variance Reduction Procedures** Procedures for reducing the error in a Monte Carlo simulation.
- Variance Swap** Swap where the realized variance rate during a period is exchanged for a fixed variance rate. Both are applied to a notional principal.
- Variation Margin** An extra margin required to bring the balance in a margin account up to the initial margin when there is a margin call.
- Vega** The rate of change in the price of an option or other derivative with volatility.

- Vega-Neutral Portfolio** A portfolio with a vega of zero.
- Vesting Period** Period during which an option cannot be exercised.
- VIX Index** Index of the volatility of the S&P 500.
- Volatility** A measure of the uncertainty of the return realized on an asset.
- Volatility Skew** A term used to describe the volatility smile when it is nonsymmetrical.
- Volatility Smile** The variation of implied volatility with strike price.
- Volatility Surface** A table showing the variation of implied volatilities with strike price and time to maturity.
- Volatility Swap** Swap where the realized volatility during a period is exchanged for a fixed volatility. Both percentage volatilities are applied to a notional principal.
- Volatility Term Structure** The variation of implied volatility with time to maturity.
- Warrant** An option issued by a company or a financial institution. Call warrants are frequently issued by companies on their own stock.
- Waterfall** Rules determining how cash flows from the underlying portfolio are distributed to tranches.
- Weather Derivative** Derivative where the payoff depends on the weather.
- Wiener Process** A stochastic process where the change in a variable during each short period of time of length Δt has a normal distribution with a mean equal to zero and a variance equal to Δt .
- Wild Card Play** The right to deliver on a futures contract at the closing price for a period of time after the close of trading.
- Writing an Option** Selling an option.
- Yield** A return provided by an instrument.
- Yield Curve** *See* Term Structure.
- Zero-Coupon Bond** A bond that provides no coupons.
- Zero-Coupon Interest Rate** The interest rate that would be earned on a bond that provides no coupons.
- Zero-Coupon Yield Curve** A plot of the zero-coupon interest rate against time to maturity.
- Zero Curve** *See* Zero-Coupon Yield Curve.
- Zero Rate** *See* Zero-Coupon Interest Rate.



DerivaGem Software

There are a number of new features of DerivaGem. The software has been simplified by eliminating the *.dll files. Source code is included with the functions, and functions are now accessible to Mac and Linux users. CDSs and CDOs can now be valued.

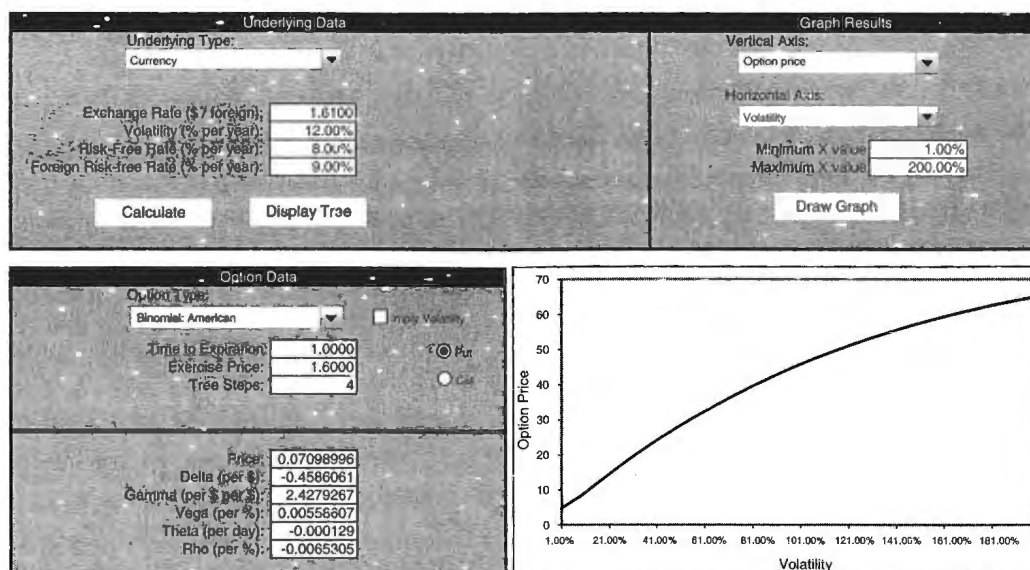
Getting Started

The most difficult part of using software is getting started. Here is a step-by-step guide to valuing an option using DerivaGem Version 2.01.

1. Put the disk that comes with this book into the CD/DVD drive on your computer. Open the Excel file DG201.xls
2. If you are using Office 2007, click on *Options* at the top of your screen (above the F column) and then click *Enable this content*. If you are not using Office 2007, make sure that the security for macros is set at medium or low. (You can do this by clicking *Tools*, followed by *Macros*, followed by *Security*.)
3. Click on the *Equity_FX_Index_Futures* worksheet tab at the bottom of the page.
4. Choose *Currency* as the Underlying Type and *Binomial American* as the Option Type. Click on the *Put* button. Leave *Implied Volatility* unchecked.
5. You are now all set to value an American put option on a currency. There are six inputs: exchange rate, volatility, domestic risk-free rate, foreign risk-free rate, time to expiration (years), exercise price, and time steps. Input these in cells D6, D7, D8, D9, D19, D20, and D21 as 1.61, 12%, 8%, 9%, 1.0, 1.60, and 4, respectively.
6. Hit *Enter* on your keyboard and click on *Calculate*. You will see the price of the option in cell D25 as 0.07099 and the Greek letters in cells D26 to D30. The screen you should have produced is shown on the following page.
7. Click on *Display Tree*. You will see the binomial tree used to calculate the option. This is Figure 20.6 in Chapter 20.

Next Steps

You should now have no difficulty valuing other types of option on other underlyings with this worksheet. To imply a volatility, check the *Implied Volatility* box and input the



option price in cell D25. Hit *Enter* and click on *Calculate*. The implied volatility is displayed in cell D7.

Many different charts can be displayed. To display a chart, you must first choose the variable you require on the vertical axis, the variable you require on the horizontal axis, and the range of values to be considered on the horizontal axis. Following that, you should hit *Enter* on your keyboard and click on *Draw Graph*.

Other points to note about this worksheet are:

1. For European and American equity options, up to 10 dividends on the underlying stock can be input in a table that pops up. Enter the time of each dividend (measured in years from today) in the first column and the amount of the dividend in the second column. Dividends must be entered in chronological order.
2. Up to 500 time steps can be used for the valuation of American options, but only a maximum of 10 time steps can be displayed.
3. Greek letters for all options other than standard calls and puts are calculated by perturbing the inputs, not by using analytic formulas.
4. For an Asian option the *Current Average* is the average price since inception. For a new deal (with zero time to inception), the current average is irrelevant.
5. In the case of a lookback option, *Minimum to Date* is used when a call is valued and *Maximum to Date* is used when a put is valued. For a new deal, these should be set equal to the current price of the underlying asset.
6. Interest rates are continuously compounded.

Bond Options

The general operation of the *Bond_Options* worksheet is similar to that of the *Equity_FX_Index_Futures* worksheet. The alternative models are Black's model (see Section 28.1), the normal model of the short rate (see equation (30.13)), and the lognormal model of the short rate (see equation (30.18)). The first model can be applied

only to European options. The other two can be applied to European or American options. The coupon is the rate paid per year and the frequency of payments can be selected as Quarterly, Semi-Annual or Annual. The zero-coupon yield curve is entered in the table labeled Term Structure. Enter maturities (measured in years) in the first column and the corresponding continuously compounded rates in the second column. The maturities must be in chronological order. DerivaGem assumes a piecewise linear zero curve similar to that in Figure 4.1. The strike price can be quoted (clean) or cash (dirty) (see Section 28.1). The quoted bond price, which is calculated by the software, and the strike price, which is input, are per \$100 of principal.

Caps and Swaptions

The general operation of the *Caps_and Swap_Options* worksheet is similar to that of the *Equity_FX_Index_Futures* worksheet. The worksheet is used to value interest rate caps/floors and swap options. Black's model for caps and floors is explained in Section 28.2 and Black's model for European swap options is explained in Section 28.3. The normal and lognormal short-rate models are in equations (30.13) and (30.18), respectively. The term structure of interest rates is entered in the same way as for bond options. The frequency of payments can be selected as Monthly, Quarterly, Semi-Annual, or Annual. The software calculates payment dates by working backward from the end of the life of the instrument. The initial accrual period for a cap/floor may be a nonstandard length between 0.5 and 1.5 times a normal accrual period.

CDSs

The CDS worksheet is used to calculate hazard rates from CDS spreads and vice versa. Users must input a term structure of interest rates (continuously compounded) and either a term structure of CDS spreads or a term structure of hazard rates. The initial hazard rate applies from time zero to the time specified; the second hazard rate applies from the time corresponding to the first hazard rate to the time corresponding to the second hazard rate; and so on. The hazard rates are continuously compounded, so that a hazard rate $h(t)$ at time t means that the probability of default between times t and $t + \Delta t$, conditional on no earlier default, is $h(t) \Delta t$. The calculations are carried out assuming that default can occur only at points midway between payment dates. This corresponds to the calculations for the example in Section 23.2 (the hazard rate in that example is 2% with annual compounding or 2.02% with continuous compounding).

CDOs

The CDO worksheet calculates quotes for the tranches of CDOs from tranche correlations input by the user. The attachment points and detachment points for tranches are input by the user. The quotes can be in basis points or involve an upfront payment. In the latter case, the spread in basis points is fixed and the upfront payment, as a percent of the tranche principal, is either input or implied. (For example, the fixed spread for the equity tranche of iTraxx Europe or CDX NA IG is 500 basis points.) The number of integration points (see equation (24.12)) defines the accuracy of calculations and can be left as 10 for most purposes (the maximum is 30). The software displays the expected loss as a percent of the tranche principal (ExpLoss) and the present value of expected

payments (PVPmts) at the rate of 10,000 basis points per year. The spread and upfront payment are

$$\text{ExpLoss} * 10,000 / \text{PVPmts} \quad \text{and} \quad \text{ExpLoss} - (\text{Spread} * \text{PVPmts} / 10,000)$$

respectively. The worksheet can be used to imply either tranche (compound) correlations or base correlations from quotes input by the user. For base correlations to be calculated, it is necessary for the first attachment point to be 0% and the detachment point for one tranche to be the attachment point for the next tranche.

How Greek Letters Are Defined

In the *Equity_FX_Index_Futures* worksheet, the Greek letters are defined as follows:

Delta: Change in option price per dollar increase in underlying asset

Gamma: Change in delta per dollar increase in underlying asset

Vega: Change in option price per 1% increase in volatility (e.g., volatility increases from 20% to 21%)

Rho: Change in option price per 1% increase in interest rate (e.g., interest increases from 5% to 6%)

Theta: Change in option price per calendar day passing.

In the *Bond_Options* and *Caps_and_Swap_Options* worksheets, the Greek letters are defined as follows:

DV01: Change in option price per 1-basis-point upward parallel shift in the zero curve

Gamma01: Change in DV01 per 1-basis-point upward parallel shift in the zero curve, multiplied by 100

Vega: Change in option price when volatility parameter increases by 1% (e.g., volatility increases from 20% to 21%).

The Applications Builder

Once you are familiar with the Options calculator (DG201.xls), you may want to start using the Application Builder. This consists of most of the functions underlying the Options Calculator with source code. It enables you to compile tables of option values, create your own charts, or develop applications. Excel users should load DG201 functions.xls and Open Office users should load Open Office DG201 functions.ods. Below are some sample applications that have been developed. They are in DG201 applications.xls and Open Office DG201 applications.ods. If any reader wishes to distribute other applications to colleagues, I would be pleased to do this (with full acknowledgements) via my website and the next release of the software.

- A. Binomial Convergence. This investigates the convergence of the binomial model in Chapters 12 and 20.
- B. Greek Letters. This provides charts showing the Greek letters in Chapter 18.
- C. Delta Hedge. This investigates the performance of delta hedging as in Tables 18.2 and 18.3.

- D.** Delta and Gamma Hedge. This investigates the performance of delta plus gamma hedging for a position in a binary option.
- E.** Value and Risk. This calculates Value at Risk for a portfolio using three different approaches.
- F.** Barrier Replication. This carries out calculations for static options replication (see Section 25.16).
- G.** Trinomial Convergence. This investigates the convergence of a trinomial tree model.

Note that E, F, and G are not included in the Open Office version of the software.

Major Exchanges Trading Futures and Options

Australian Securities Exchange (ASX)	www.asx.com.au
BM&FBOVESPA (BMF)	www.bmfbovespa.com.br
Bombay Stock Exchange (BSE)	www.bseindia.com
Boston Options Exchange (BOX)	www.bostonoptions.com
Bursa Malaysia (BM)	www.bursamalaysia.com
Chicago Board Options Exchange (CBOE)	www.cboc.com
China Financial Futures Exchange (CFFEX)	www.cffex.com.cn
CME Group	www.cmegroup.com
Dalian Commodity Exchange (DCE)	www.dce.com.cn
Eurex	www.eurexexchange.com
Hong Kong Futures Exchange (HKFE)	www.hkex.com.hk
IntercontinentalExchange (ICE)	www.theice.com
International Securities Exchange (ISE)	www.iseoptions.com
Kansas City Board of Trade (KCBT)	www.kcbt.com
London Metal Exchange (LME)	www.lme.co.uk
MEFF Renta Fija and Variable, Spain	www.meff.es
Mexican Derivatives Exchange (MEXDER)	www.mexder.com
Minneapolis Grain Exchange (MGE)	www.mgex.com
Montreal Exchange (ME)	www.m-x.ca
NASDAQ OMX	www.nasdaqomx.com
National Stock Exchange, Mumbai (NSE)	www.nseindia.com
NYSE Euronext	www.nyse.com
Osaka Securities Exchange (OSE)	www.ose.or.jp
Shanghai Futures Exchange (SHFE)	www.shfe.com.cn
Singapore Exchange (SGX)	www.sgx.com
Tokyo Grain Exchange (TGE)	www.tge.or.jp
Tokyo Financial Exchange (TFX)	www.tfx.co.jp
Zhengzhou Commodity Exchange (ZCE)	www.zce.cn

There has been a great deal of consolidation of derivatives exchanges, nationally and internationally, in the last few years. The Chicago Board of Trade and the Chicago Mercantile Exchange have merged to form the CME Group, which also includes the New York Mercantile Exchange (NYMEX). Euronext and the NYSE have merged to form NYSE Euronext, which now owns the American Stock Exchange (AMEX), the Pacific Exchange (PXS), the London International Financial Futures Exchange (LIFFE), and two French exchanges. The Australian Stock Exchange and the Sydney Futures Exchange (SFE) have merged to form the Australian Securities Exchange (ASX). The IntercontinentalExchange (ICE) has acquired the New York Board of Trade (NYBOT), the International Petroleum Exchange (IPE), and the Winnipeg Commodity Exchange (WCE). Eurex, which is jointly operated by Deutsche Borse AG and SIX Swiss Exchange, has acquired the International Securities Exchange (ISE). No doubt the consolidation has been largely driven by economies of scale that lead to lower trading costs.

Table for $N(x)$ When $x \leq 0$

This table shows values of $N(x)$ for $x \leq 0$. The table should be used with interpolation. For example,

$$\begin{aligned} N(-0.1234) &= N(-0.12) - 0.34[N(-0.12) - N(-0.13)] \\ &= 0.4522 - 0.34 \times (0.4522 - 0.4483) \\ &= 0.4509 \end{aligned}$$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-3.0	0.0014	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
-3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
-3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
-4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table for $N(x)$ When $x \geq 0$

This table shows values of $N(x)$ for $x \geq 0$. The table should be used with interpolation. For example,

$$\begin{aligned} N(0.6278) &= N(0.62) + 0.78[N(0.63) - N(0.62)] \\ &= 0.7324 + 0.78 \times (0.7357 - 0.7324) \\ &= 0.7350 \end{aligned}$$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9986	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4.0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000



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